

Ricky Kumar

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#)

Summary

M.Tech Artificial Intelligence student focused on building agentic and RAG-based LLM systems that automate decision-making and reduce hallucination. Experienced in evaluating LLM outputs with LangSmith, optimizing fine-tuning (PEFT, QLoRA, transfer learning) to cut training cost, and orchestrating multi-agent, multilingual (text, voice, video) pipelines for enterprises handling sensitive data and policy documents.

Education

IIIT Pune

Aug 2025 – May 2027

M.Tech in Artificial Intelligence, CGPA: 7.81

Coursework: Gen-AI, Agentic AI, Deep Learning, Machine Learning, Big Data Analytics, Cryptocurrency & Blockchain, Research Methodology

MAIT Solan

Aug 2018 – May 2022

B.Tech in Computer Science, CGPA: 7.7

Coursework: Data Structures & Algorithms, C/C++, Python, Web Development, ML/DL, OOPs, COA, CN, CD, OS, TOC, Linux, DM, SE, Distributed Computing, Soft Computing

Projects

MultiAgent AI Interview Platform

Aug 2026 – Present

- Architected a multi-agent AI system for mock interviews, resume building, and self-evaluation, using **React**, **Node.js**, **Express**, and **FastAPI** across a microservices-based backend.
 - Implemented secure authentication and session management with **Firestore** and **Google OAuth**, and containerized services using **Docker** for scalable, isolated deployments.
 - Designed a caching and session layer with **Redis** and a persistent data store with **MongoDB** to handle real-time interview state and user progress tracking.
 - Integrated a secure payment gateway for premium features and built an API gateway to route requests across independent AI agent services.
- Multi-Agent Hotel Booking Assistant – Taj Aguada (AWS Bedrock)** Jun – Jul 2026

- Architected a multi-agent booking system on Amazon Bedrock Agents, orchestrating sub-agents for room search, availability, booking confirmation, and guest FAQs.
- Built a Bedrock Knowledge Base (Titan Embeddings, ~300-token chunks, 15% overlap) over policies, tariffs, and amenities for accurate retrieval.
- Used Claude on Bedrock as the orchestrator to parse intent, trigger Lambda action groups for live availability/booking, and generate responses; persisted session and booking state in DynamoDB.
- Benchmarked Claude/Titan token pricing and tuned chunk size and top-k to optimize answer quality per API cost.
- Applied Bedrock Guardrails for PII masking, denied-topic filtering, and knowledge-grounded responses to eliminate hallucinated booking details.

UPSC Study Assistant – Corrective RAG with Local LLM/LLM API

Mar – May 2026

- Ingested study material (PDFs via PyPDF2, lecture audio via Whisper, figures via CLIP, and diagrams via OCR) into a FAISS vector store for unified multimodal retrieval.
- Built a corrective RAG pipeline in which a local open-source Llama model scores each retrieved chunk for relevance, discards low-quality results, and reranks the remaining documents before generation.
- Used Redis for short-term session memory and PostgreSQL to track long-term learning patterns and user habits, enabling personalized study recommendations.

Phishing Detection Model

Aug – Dec 2025

- Built a layer that intercepts suspicious web links, email links, and executes them in an isolated Selenium sandbox before they reach the user's browser.
- Detected fake login forms, hidden password fields, and malicious API calls via real-time DOM and JavaScript behavioral analysis, blocking credential harvesting.
- Performed data pipeline tasks including preprocessing, cleaning, modeling, and deployment.

Technical Skills

Languages:	C, C++, Python, javascript, SQL
Databases & Retrieval:	MySQL, MongoDB, DynamoDB, FAISS, PostgreSQL, Redis
ML/DL Tools:	PyTorch, NumPy, Pandas, scikit-learn, Matplotlib, Jupyter
DL Architectures:	CNN, RNN, ANN, Encoder-Decoder, U-Net
Models:	GAN, VAE, Random Forest, Logistic/Linear Regression, KNN, DBSCAN, Decision Tree, SVM
Generative AI & LLM:	LangGraph, LangChain, Self-Corrective RAG, Fine-tuning, Ollama, Whisper, CLIP, Prompt Engineering, Guardrails
Web dev & Deployment:	React, Node.js, Express, FastAPI, MCP, AWS SageMaker, Docker, Firebase, Google Auth
Safety & Monitoring:	MLOps, Bias Detection, Compliance Checks, Prompt Injection Prevention
Tools:	Spread sheet, Power BI, Git, Github, Colab, kaggle, Jupyter, Cursor, Vscode, Copilot.

Certifications

- Agentic AI on AWS (Bedrock AgentsCore) – Udemy
- Complete AI, Machine Learning & Data Science – Udemy
- Google Advance Data analytics – Coursera
- CCNA Computer network from – GICSEH, noida
- Python programming from – csdtit, patna

Experience & Achievements

- Teaching Assistant, IIIT Pune** – Lab evaluation and mentoring for OOPs, DSA, DBMS, Cloud Computing, and Operating Systems
- GATE 2025:** All India Rank ~10,000 out of 170,000 in Computer Science (Score: 448/1000)
- LeetCode:** 200+ problems solved across data structures and algorithms